

A Spectral Method for Identifiable Grade of Membership Analysis with Binary Responses

Abstract

Grade of Membership (GoM) models are popular individual-level mixture models for multivariate categorical data. GoM allows each subject to have mixed memberships in multiple extreme latent profiles. Therefore GoM models have a richer modeling capacity than latent class models that restrict each subject to belong to a single profile. The flexibility of GoM comes at the cost of more challenging identifiability and estimation problems. In this work, we propose a singular value decomposition (SVD) based spectral approach to GoM analysis with multivariate binary responses. Our approach hinges on the observation that the expectation of the data matrix has a low-rank decomposition under a GoM model. For *identifiability*, we develop sufficient and almost necessary conditions for a notion of expectation identifiability. For *estimation*, we extract only a few leading singular vectors of the observed data matrix, and exploit the simplex geometry of these vectors to estimate the mixed membership scores and other parameters. Our spectral method has a huge computational advantage over Bayesian or likelihood-based methods and is scalable to large-scale and high-dimensional data. Extensive simulation studies demonstrate the superior efficiency and accuracy of our method. We also illustrate our method by applying it to a personality test dataset.

Keywords: Grade of Membership Model; Identifiability; Latent Variable Model; Mixed Membership Model; Sequential Projection Algorithm; Singular Value Decomposition; Spectral Method.

1 Introduction

Multivariate categorical data are routinely collected in various social and behavioral sciences, such as psychological tests (Chen et al., 2019), educational assessments (Shang et al., 2021), and political surveys (Chen et al., 2021b). In these applications, it is often of great interest to use latent variables to model the unobserved constructs such as personalities, abilities, political ideologies, etc. Popular latent variable models for multivariate categorical

data include the item response theory models (IRT; [Embretson and Reise, 2013](#)) and latent class models (LCM; [Hagenaars and McCutcheon, 2002](#)), which employ continuous and discrete latent variables, respectively. Different from these modeling approaches, the grade of membership (GoM) models ([Woodbury et al., 1978](#); [Erosheva, 2002, 2005](#)) allow each observation to have mixed memberships in multiple *extreme latent profiles*. GoM models assume that each observation has a latent membership vector with K continuous membership scores that sum up to one. Each membership score quantifies the extent to which this observation belongs to each of K extreme profiles. So GoM can be viewed as incorporating both the continuous aspect (via the membership scores) and discrete aspect (via the K extreme latent profiles) of latent variables. More generally, GoM belongs to the broad family of mixed membership models for individual-level mixtures ([Airoldi et al., 2014](#)). Thanks to their nice interpretability and rich expressive power, variants of mixed membership models including GoM are widely used in many applications such as survey data modeling ([Erosheva et al., 2007](#)), response time modeling ([Pokropek, 2016](#)), topic modeling ([Blei et al., 2003](#)), social networks ([Airoldi et al., 2008](#)), and data privacy ([Manrique-Vallier and Reiter, 2012](#)).

The flexibility of GoM models comes at the cost of more challenging identifiability and estimation problems. In the existing literature on GoM model estimation, Bayesian inference using Markov Chain Monte Carlo (MCMC) is perhaps the most prevailing approach ([Erosheva, 2002](#); [Erosheva et al., 2007](#); [Gormley and Murphy, 2009](#); [Gu et al., 2023](#)). However, the posterior distributions of the GoM model parameters are complicated, after integrating out the individual-level membership scores. Many studies developed advanced MCMC algorithms for approximate posterior computation. Yet MCMC sampling is time-consuming and typically not computationally efficient. On the other hand, the frequentist estimation approach of marginal maximum likelihood (MML) would bring a similar challenge, because the marginal likelihood still involves the intractable integrals of those latent membership scores.

Actually, it was pointed out in [Borsboom et al. \(2016\)](#) that GoM models are very useful in identifying meaningful profiles in applications including depression, personality disorders, etc., but they are temporarily not widely used in psychometrics due to the lack of readily accessible and efficient statistical software.

Recently, the development of the R package `sirt` ([Robitzsch and Robitzsch, 2022](#)) provides a joint maximum likelihood (JML) algorithm for GoMs based on the iterative estimation method proposed in [Erosheva \(2002\)](#). In contrast to MML, the JML approach treats the subjects' latent membership scores as fixed unknown parameters rather than random quantities. This approach hence circumvents the need of evaluating the intractable integrals during estimation. JML is currently considered the most efficient tool for estimating GoM models. However, due to its iterative manner, JML's efficiency is still unsatisfactory when applied to very large-scale data with many observations and many items. Therefore, it is desirable to develop more scalable and non-iterative estimation methods and aid psychometric researchers and practitioners to perform GoM analysis of modern item response data.

In addition to the difficulty of estimation, model identifiability is also a challenging issue for GoM models. A model is identifiable if the model parameters can be reliably recovered from the observed data. Identifiability is crucial to ensuring valid statistical estimation as well as meaningful interpretation of the inferred latent structures. The handbook of [Airoldi et al. \(2014\)](#) emphasizes theoretical difficulties of identifiability in mixed membership models, including GoM models. Recently, recognizing the difficulty of establishing identifiability of GoM models, [Gu et al. \(2023\)](#) proposed to incorporate a dimension-grouping modeling component to GoM and established the population identifiability for this new model. However, their identifiability results do not apply to the original GoM. In addition, their identifiability notion only concerns the population parameters in the model, but excludes the individual-level latent membership scores.

To address the aforementioned issues, we propose a novel singular value decomposition (SVD) based spectral approach to GoM analysis with multivariate binary data. Our approach hinges on the observation that the expectation of the response matrix admits a low-rank decomposition under GoM. Our contributions are two-fold. First, we consider a notion of *expectation identifiability* and establish identifiability for GoM models with binary responses. Under this new notion, the identifiable quantities include not only the population parameters, but also the individual membership scores that indicate the grades of memberships. Specifically, we derive sufficient conditions that are almost necessary for identifiability. Second, based on our new identifiability results, we propose an SVD-based spectral estimation method scalable to large-scale and high-dimensional data. In the simulation studies, we empirically verify the identifiability results and also demonstrate the superior efficiency and accuracy of our proposed algorithm. A real data example also illustrates that meaningful interpretation can be drawn from the results of our proposed method.

The rest of the paper is structured as follows. Section 2 introduces the model setup and lays out the motivation for this work. Section 3 presents the identifiability results. Section 4 proposes a spectral estimation algorithm. Section 5 conducts simulation studies to assess the performance of the proposed method and empirically verify the identifiability results. Section 6 illustrates the proposed method using a real data example in a psychological test. Finally, Section 7 concludes the paper and discusses future research directions. The proofs of the identifiability results are included in the Appendix.

2 Model Setup and Motivation

GoM models can be used to model multivariate categorical data with a mixed membership structure. In this work, we focus on multivariate binary responses, which are very commonly

encountered in social, behavioral, and biomedical applications, including yes/no responses in social science surveys (Erosheva et al., 2007; Chen et al., 2021b), correct/wrong answers in educational assessments (Shang et al., 2021), and presence/absence of symptoms in medical diagnosis (Woodbury et al., 1978). We point out that our identifiability results and spectral estimation method in the binary case will illuminate the key structure of GoM and pave the way for generalizing to the general categorical response case. We will briefly discuss the possibility of such extensions in Section 7. In our binary response setting, denote the number of items by J . For a random subject i , denote his or her observed response to the j -th item by $R_{ij} \in \{0, 1\}$ for $j = 1, \dots, J$.

A GoM model is characterized by two levels of modeling: the population level and the individual level. On the population level, K *extreme latent profiles* are defined to capture a finite number of prototypical response patterns. For $k \in [K]$, the k -th extreme latent profile is characterized by the item parameter vector $\boldsymbol{\theta}_k = (\theta_{1k}, \dots, \theta_{JK})$ with $\theta_{jk} \in [0, 1]$ collecting the Bernoulli parameters of conditional response probabilities. Specifically,

$$\theta_{jk} = \mathbb{P}(R_{ij} = 1 \mid \text{subject } i \text{ solely belongs to the } k\text{-th extreme latent profile}). \quad (1)$$

We collect all the item-level Bernoulli parameters in a $J \times K$ matrix $\boldsymbol{\Theta} = (\theta_{jk}) \in \mathbb{R}^{J \times K}$. On the individual level, each subject i has a latent membership vector $\boldsymbol{\pi}_i = (\pi_{i1}, \dots, \pi_{iK})$, satisfying $\pi_{ik} \geq 0$ and $\sum_{k=1}^K \pi_{ik} = 1$. Here π_{ik} indicates the extent to which subject i partially belongs to the k -th extreme profile, and they are called membership scores. It is now instrumental to compare the assumption of the GoM model with that of the latent class model (LCM; Goodman, 1974; Hagenaars and McCutcheon, 2002). Both GoM and LCM share a similar formulation of the item parameters $\boldsymbol{\Theta}$ as defined in (1). However, under an LCM, each subject i is associated with a categorical variable $z_i \in [K]$ instead

of a membership vector. This means LCM restricts each subject to solely belonging to a single profile, as opposed to partially belonging to multiple profiles in the GoM. Further, the fundamental representation theorem in [Erosheva et al. \(2007\)](#) shows that a GoM model can be reformulated with an LCM but with K^J latent classes instead of K latent classes. In summary, GoM is a more general and flexible tool than LCM for modeling multivariate data, but also exhibits a more complicated model structure. It is therefore more challenging to establish identifiability and perform estimation under the GoM setting.

Given the membership score vector $\boldsymbol{\pi}_i$ and the item parameters $\boldsymbol{\Theta}$, the conditional probability of the i -th subject providing a positive response to the j -th item is

$$\mathbb{P}(R_{ij} = 1 \mid \boldsymbol{\pi}_i, \boldsymbol{\Theta}) = \sum_{k=1}^K \pi_{ik} \theta_{jk}. \quad (2)$$

In other words, the positive response probability for a subject to an item is a convex combination of the extreme profile response probabilities θ_{jk} weighted by the subject's membership scores π_{ik} . The GoM model assumes that a subject's responses $R_{i,1}, \dots, R_{i,J}$ are conditionally independent given the membership scores $\boldsymbol{\pi}_i$. We consider a sample of N i.i.d. subjects, and collect all the membership scores in a $N \times K$ matrix $\boldsymbol{\Pi} = (\pi_{ik}) \in \mathbb{R}^{N \times K}$.

In the GoM modeling literature (e.g. [Erosheva, 2002](#)), there are two perspectives of dealing with the membership scores $\boldsymbol{\pi}_i$: the random-effect and the fixed-effect perspectives. The random effect perspective treats $\boldsymbol{\pi}_i$ as random and assumes that they follow some distribution parameterized by $\boldsymbol{\alpha}$: $\boldsymbol{\pi}_i \sim D_{\boldsymbol{\alpha}}(\cdot)$. Note that $\boldsymbol{\pi}_i \in \Delta_{K-1} = \{\mathbf{x} = (x_1, \dots, x_K) : x_i \geq 0, \sum_{i=1}^K x_i = 1\}$ where Δ_{K-1} denotes the probability simplex. A common choice of the distribution $D_{\boldsymbol{\alpha}}$ for $\boldsymbol{\pi}_i$ is the Dirichlet distribution ([Blei et al., 2003](#)).

From the above random-effect perspective, the *marginal likelihood* function for a GoM

model is

$$L(\boldsymbol{\Theta}, \boldsymbol{\alpha} \mid \mathbf{R}) = \prod_{i=1}^N \int_{\Delta_{K-1}} \prod_{j=1}^J \left(\sum_{k=1}^K \pi_{ik} \theta_{jk} \right)^{R_{ij}} \left(1 - \sum_{k=1}^K \pi_{ik} \theta_{jk} \right)^{1-R_{ij}} dD_{\boldsymbol{\alpha}}(\boldsymbol{\pi}_i), \quad (3)$$

where the membership scores $\boldsymbol{\pi}_i$'s are marginalized out with respect to their distribution $D_{\boldsymbol{\alpha}}$. The integration of the products of sums in (3) poses challenges to establishing identifiability. This difficulty motivated Gu et al. (2023) to introduce a dimension-grouping component to simplify the integrals and then prove identifiability for that new model. In terms of estimation, the marginal maximum likelihood (MML) approach maximizes (3) to estimate the population parameters $(\boldsymbol{\Theta}, \boldsymbol{\alpha})$ rather than the individual membership scores $\boldsymbol{\Pi}$. Bayesian inference with MCMC is also often used for estimation (Erosheva et al., 2007; Manrique-Vallier and Reiter, 2012; Gu et al., 2023), where inferring parameters like $\boldsymbol{\alpha}$ in the Dirichlet distribution $D_{\boldsymbol{\alpha}}(\cdot)$ typically require the Metropolis-Hastings sampling.

On the other hand, the fixed-effect perspective of GoM treats the membership scores $\boldsymbol{\pi}_i$ as fixed unknown parameters and aims to directly estimate them. This approach does not model the distribution of the membership scores, and hence circumvents the need of evaluating the intractable integrals during estimation. In this case, if still adopting the likelihood framework, the *joint likelihood* function of both $\boldsymbol{\Theta}$ and $\boldsymbol{\Pi}$ for a GoM model is

$$L(\boldsymbol{\Pi}, \boldsymbol{\Theta} \mid \mathbf{R}) = \prod_{i=1}^N \prod_{j=1}^J \left(\sum_{k=1}^K \pi_{ik} \theta_{jk} \right)^{R_{ij}} \left(1 - \sum_{k=1}^K \pi_{ik} \theta_{jk} \right)^{1-R_{ij}}. \quad (4)$$

The joint maximum likelihood (JML) approach maximizes (4) to estimate $\boldsymbol{\Theta}$ and $\boldsymbol{\Pi}$. Based on an iterative algorithm proposed by Erosheva (2002), the R package `sirt` (Robitzsch and Robitzsch, 2022) provides a function `JML` to solve this optimization problem under GoM. JML methods are typically known as inconsistent for many traditional models (Neyman and Scott, 1948) when the sample size goes to infinity (large N) but the number of observed

variables is finite (fixed J). Nonetheless, in modern large-scale assessments or surveys where the data collection scope is unprecedentedly big and high-dimensional, both N and J can be quite large. JML is currently considered the most efficient tool for estimating GoM models. However, due to its iterative manner, JML's efficiency is still unsatisfactory when applied to modern big datasets with many observations and many items. Therefore, it is desirable to develop more scalable and non-iterative estimation methods and aid psychometric researchers and practitioners to perform GoM analysis of item response data.

To address the above issues in the GoM analysis, in this work, we propose a novel singular value decomposition (SVD) based spectral approach. Our approach hinges on the observation that the expectation of the response matrix under a GoM model admits a low-rank decomposition. To see this, it is useful to summarize (2) in matrix form

$$\mathbf{R}_0 := \mathbb{E}[\mathbf{R}] = \underbrace{\mathbf{\Pi}}_{N \times K} \underbrace{\mathbf{\Theta}^\top}_{K \times J}, \quad (5)$$

where $\mathbf{R} = (R_{ij}) \in \{0, 1\}^{N \times J}$ denotes the binary response data matrix, and $\mathbf{R}_0 = (R_{0,ij}) \in \mathbb{R}^{N \times J}$ is the element-wise expectation of $\mathbf{R}$. Note that the factorization in (5) implies that the $N \times J$ matrix $\mathbf{R}_0$ has rank at most K , which is the number of extreme latent profiles. Since K is typically (much) smaller than N and J , the decomposition (5) exhibits a low-rank structure. Therefore, we can consider the singular value decomposition (SVD) of $\mathbf{R}_0$:

$$\mathbf{R}_0 = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top, \quad (6)$$

where $\mathbf{\Sigma}$ is a $K \times K$ diagonal matrix collecting the K singular values of $\mathbf{R}_0$; denote these singular values by $\sigma_1 \geq \dots \geq \sigma_K \geq 0$ and write $\mathbf{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_K)$. Matrices $\mathbf{U}_{N \times K}$, $\mathbf{V}_{J \times K}$ collect the corresponding left and right singular vectors and satisfy $\mathbf{U}^\top \mathbf{U} = \mathbf{V}^\top \mathbf{V} = \mathbf{I}_K$. Our high-level idea is to utilize the top K left singular vectors of the data matrix $\mathbf{R}$ to

identify and estimate $\mathbf{\Pi}$ and subsequently, $\mathbf{\Theta}$. In the following two sections, we will present new identifiability results and develop a spectral estimation algorithm for GoM models based on the SVD in (6).

3 Identifiability Results

The study of identifiability in statistics dates back to [Koopmans and Reiersol \(1950\)](#). A model is identifiable if the model parameters can be reliably recovered from the observed data. Identifiability is a crucial property of a statistical model as it is a prerequisite for valid and reproducible statistical inference. In latent variable modeling, identifiability is especially essential since it is a foundation for meaningful interpretation of the latent constructs.

Traditionally, identifiability of a statistical model means that the population parameters can be uniquely determined from the marginal distribution of the observed variables ([Koopmans and Reiersol, 1950](#); [Goodman, 1974](#)). In the context of GoM models, this notion of *population identifiability* is equivalent to identifying parameters $(\mathbf{\Theta}, \mathbf{\alpha})$ from the marginal distribution in (3). The complicated integrals in (3) make it difficult to establish population identifiability, which motivated [Gu et al. \(2023\)](#) to propose a dimension-grouping modeling component to simplify GoM and prove identifiability for that new model. However, it remains unknown whether the original GoM models can be identified.

In this work, we consider a new notion of identifiability, which we term as *expectation identifiability*. This notion concerns not only the item parameters, but also the individual membership scores. Similar identifiability notions are widely adopted and studied in the network modeling and topic modeling literature, e.g., [Mao et al. \(2021\)](#) and [Ke and Wang \(2022\)](#). Specifically, recall from (5) that the expectation of the data matrix $\mathbf{R}$ has a low-rank decomposition $\mathbf{R}_0 = \mathbf{\Pi}\mathbf{\Theta}^\top$, we seek to understand under what conditions this decomposition

is unique. Note that both the Bernoulli probabilities Θ and the membership scores Π are treated as parameters to be identified. We call a parameter set (Π, Θ) valid if $\pi_i \in \Delta_{K-1}$ and $\theta_{jk} \in [0, 1]$ for all i, j , and k . We formally define expectation identifiability below.

Definition 1 (Expectation identifiability). *A GoM model with parameter set (Π, Θ) is said to be identifiable, if for any other valid parameter set $(\tilde{\Pi}, \tilde{\Theta})$, $\tilde{\Pi}\tilde{\Theta}^\top = \Pi\Theta^\top$ holds if and only if (Π, Θ) and $(\tilde{\Pi}, \tilde{\Theta})$ are identical up to a permutation of the K extreme profiles.*

One might ask whether identifying Π and Θ from the expectation $\mathbf{R}_0$ has any implications on identifying Π and Θ from the observed data $\mathbf{R}$. In fact, when both N and J are large with respect to K , it is known that the difference between the low-rank decompositions of $\mathbf{R}_0$ and $\mathbf{R}$ is small in a certain sense (Chen et al., 2021a). We will revisit and elaborate on this subtlety when describing our estimation method and presenting the simulation results. Briefly speaking, studying the expectation identifiability problem from $\mathbf{R}_0$ is very meaningful in modern large-scale and high-dimensional data settings with large N and J .

We next present our new identifiability conditions for GoM models. We first define the important concept of pure subjects.

Definition 2 (Pure subject). *Subject i is a pure subject for extreme profile k if the only positive entry of π_i is located at index k ; that is,*

$$\pi_i = (0, \dots, 0, \underbrace{1}_{k\text{-th entry}}, 0, \dots, 0).$$

In words, subject i is a pure subject for profile k if it solely belongs to this profile and has no membership in any other profile. We consider the following condition for Π .

Condition 1. *Π satisfies that every extreme latent profile has at least one pure subject.*

Condition 1 is a quite mild assumption on Π , because it only requires that each of the

K extreme profiles has at least one representative subject among all N subjects. Intuitively, this condition is reasonable because the existence of these representative subjects indeed helps pinpoint the meaning and interpretation of the extreme profiles. Under Condition 1, $\mathbf{\Pi}$ contains one identity submatrix $\mathbf{I}_K$ after some row permutation. For any matrix $\mathbf{A}$, we use $\mathbf{A}_{\mathbf{S},:}$ to denote the rows with indices in $\mathbf{S}$. Under Condition 1, denote $\mathbf{S} = (S_1, \dots, S_K)$ as the index vector of one set of K pure subjects such that $\mathbf{\Pi}_{\mathbf{S},:} = \mathbf{I}_K$. So $S_1, \dots, S_K$ are distinct integers ranging in $\{1, \dots, N\}$. For example, if the first K rows of $\mathbf{\Pi}$ is equal to $\mathbf{I}_K$, then $\mathbf{S} = (1, 2, \dots, K)$. Recall $\mathbf{R}_0 = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ is the SVD for $\mathbf{R}_0$, where $\mathbf{U}$ is a $N \times K$ matrix collecting the K left singular vectors as columns. Interestingly, Condition 1 induces a *simplex geometry* on the row vectors of $\mathbf{U}$. We have the following important proposition, which serves as a foundation for both our identifiability results and estimation procedure.

Proposition 1. *Under Condition 1, the left singular matrix $\mathbf{U}$ satisfies*

$$\mathbf{U} = \mathbf{\Pi}\mathbf{U}_{\mathbf{S},:}. \quad (7)$$

Furthermore, $\mathbf{\Pi}$ and $\mathbf{\Theta}$ can be written as

$$\mathbf{\Pi} = \mathbf{U}\mathbf{U}_{\mathbf{S},:}^{-1}, \quad (8)$$

$$\mathbf{\Theta} = \mathbf{V}\mathbf{\Sigma}\mathbf{U}^\top\mathbf{\Pi}(\mathbf{\Pi}^\top\mathbf{\Pi})^{-1} = \mathbf{V}\mathbf{\Sigma}\mathbf{U}_{\mathbf{S},:}^\top. \quad (9)$$

We elaborate more on Proposition 1 below. Equation (7) in Proposition 1 implies that the left singular matrix $\mathbf{U}$ and the membership score matrix $\mathbf{\Pi}$ differ by a linear transformation, which is the $K \times K$ matrix $\mathbf{U}_{\mathbf{S},:}$. Equations (8) and (9) imply that if an index set of pure subjects $\mathbf{S}$ is known, then the parameters of interest $\mathbf{\Pi}$ and $\mathbf{\Theta}$ can be written in closed forms

in terms of the SVD and $\mathbf{S}$. More specifically, (7) is equivalent to

$$\mathbf{U}_{i,:} = \sum_{k=1}^K \pi_{ik} \mathbf{U}_{S_k,:}, \quad i = 1, \dots, N. \quad (10)$$

Each $\mathbf{U}_{i,:}$ is the embedding of the i -th subject into the top- K left singular subspace of $\mathbf{R}_0$, and all the rows $\mathbf{U}_{1,:}, \dots, \mathbf{U}_{N,:}$ can be plotted as points in the K -dimensional Euclidean space $\mathbb{R}^K$. Geometrically, since $\sum_{k=1}^K \pi_{ik} = 1$ with $\pi_{ik} \geq 0$, we know that each $\mathbf{U}_{i,:}$ is a convex combination of $\mathbf{U}_{S_1,:}, \dots, \mathbf{U}_{S_K,:}$, which are the embeddings of the K types of pure subjects. This means in $\mathbb{R}^K$, all the subjects lie in a simplex (i.e., the generalization of a triangle or tetrahedron to higher dimensions) whose vertices are these K types of pure subjects. Note that $\mathbf{U}_i$ and $\mathbf{U}_{i'}$ overlap if they have the same membership score vectors $\boldsymbol{\pi}_i = \boldsymbol{\pi}_{i'}$. Figure 1 gives an illustration of this simplex geometry on such embeddings in $\mathbb{R}^3$ with $K = 3$.

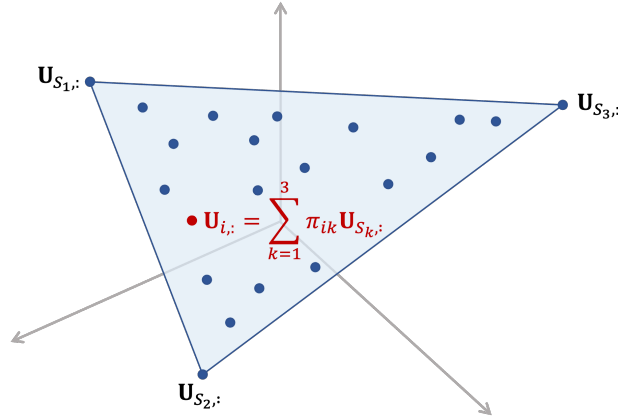


Figure 1: Illustration of the simplex geometry of the $N \times K$ left singular matrix $\mathbf{U}$ with $K = 3$. The solid dots represent the row vectors of $\mathbf{U}$ in $\mathbb{R}^3$, and the three simplex vertices (i.e, vertices of the triangle) correspond to the three types of pure subjects. All the dots lie in this triangle.

We next present our identifiability results for GoM models. We first show that the pure-subject Condition 1 is almost necessary for the identifiability of a GoM model.

Theorem 1. *Suppose $\theta_{jk} \in (0, 1)$ for all $j = 1, \dots, J$ and $k = 1, \dots, K$. If there is one*

extreme profile that does not have any pure subject, then the GoM model is not identifiable.

The proofs of the theorems are all deferred to the Appendix. Theorem 1 reveals the importance of Condition 1 for identifiability. In fact, later we will use this condition as a foundation for our estimation algorithm. Our next theorem presents sufficient and almost necessary conditions for GoM models to be identifiable.

Theorem 2. *Suppose Π satisfies Condition 1.*

- (a) *If $\text{rank}(\Theta) = K$, then the GoM model is identifiable.*
- (b) *If $\text{rank}(\Theta) = K - 1$ and no column of Θ is an affine combination of the other columns of Θ , then the GoM model is identifiable. (An affine combination of vectors $\mathbf{x}_1, \dots, \mathbf{x}_n$ is defined as $\sum_{i=1}^n a_i \mathbf{x}_i$ with $\sum_{i=1}^n a_i = 1$.)*
- (c) *In any other case, if there exists a subject i such that $\pi_{ik} > 0$ for every $k = 1, \dots, K$, then the GoM model is not identifiable.*

The high-level proof idea of Theorem 2 shares a similar spirit to Theorem 2.1 in Mao et al. (2021). We next explain and interpret the three settings in Theorem 2. According to part (a), if the K item parameter vectors $\theta_1, \dots, \theta_K$ (i.e., the K columns of Θ) are linearly independent, then the GoM model is identifiable under Condition 1. In part (b), for identifiability to hold when $\text{rank}(\Theta) = K - 1$, any item parameter vector θ_k cannot be written as an affine combination of the remaining vectors $\{\theta_{k'} : k' \neq k\}$. This is a weaker requirement on Θ compared to part (a). Part (c) states that if the conditions in parts (a) or (b) do not hold and if there exists a *completely mixed subject* that partially belongs to all profiles (i.e., $\pi_{ik} > 0$ for all k), then the model is not identifiable. Part (c) also shows that the sufficient identifiability conditions in (a) and (b) are close to being necessary, because the existence of a completely mixed subject is a very mild assumption. We next further give three toy examples with $K = 3$ and $J = 4$ to illustrate the conditions in Theorem 2.

Example 1. Consider

$$\mathbf{\Theta} = \begin{pmatrix} 0.2 & 0.8 & 0.8 \\ 0.2 & 0.8 & 0.2 \\ 0.8 & 0.2 & 0.8 \\ 0.8 & 0.2 & 0.2 \end{pmatrix}.$$

It is easy to verify that $\text{rank}(\mathbf{\Theta}) = K = 3$. This case falls into scenario (a) in Theorem 2, so a GoM model parameterized by $(\mathbf{\Pi}, \mathbf{\Theta})$ is identifiable if $\mathbf{\Pi}$ satisfies Condition 1.

Example 2. Consider

$$\mathbf{\Theta} = \begin{pmatrix} 0.2 & 0.8 & 0.8 \\ 0.2 & 0.8 & 0.8 \\ 0.8 & 0.2 & 0.8 \\ 0.8 & 0.2 & 0.8 \end{pmatrix}.$$

Now $\text{rank}(\mathbf{\Theta}) = K - 1 = 2$ since the third column of $\mathbf{\Theta}$ is a linear combination of the first two columns. However, there is no column of $\mathbf{\Theta}$ that is an affine combination of the other columns of $\mathbf{\Theta}$. This case falls into scenario (b) in Theorem 2, so a GoM model parameterized by $(\mathbf{\Pi}, \mathbf{\Theta})$ is identifiable if $\mathbf{\Pi}$ satisfies Condition 1.

Example 3. Consider

$$\mathbf{\Theta} = \begin{pmatrix} 0.2 & 0.8 & 0.5 \\ 0.2 & 0.8 & 0.5 \\ 0.8 & 0.2 & 0.5 \\ 0.8 & 0.2 & 0.5 \end{pmatrix}.$$

In this case, $\text{rank}(\mathbf{\Theta}) = K - 1 = 2$, and the third column of $\mathbf{\Theta}$ is an affine combination of the first two columns. This case falls into scenario (c) in Theorem 2, so if there exists a subject that partially belongs to all K profiles, then the GoM model is not identifiable.

4 An SVD-based Spectral Estimation Method

In the literature of GoM model estimation, the most prevailing approaches are perhaps Bayesian inferences using Markov chain Monte Carlo (MCMC) algorithms such as Gibbs and Metropolis-Hastings sampling (Erosheva, 2002; Erosheva et al., 2007; Gu et al., 2023). However, MCMC is time-consuming and typically not computationally efficient. As Borsboom et al. (2016) points out, despite their usefulness, GoM models are somewhat underrepresented in psychometric applications due to the lack of readily accessible statistical software. Recently, the R package `sirt` (Robitzsch and Robitzsch, 2022) provides a joint maximum likelihood (JML) algorithm to fit GoM models. This algorithm implements the Lagrange multiplier method proposed in Erosheva (2002) and solves the optimization problem in a gradient descent fashion. Although this JML algorithm is computationally more efficient compared to MCMC algorithms, it is still not scalable to very large-scale response data due to its iterative manner. Therefore, it is of interest to develop a non-iterative estimation method suitable to analyze modern datasets with a large number of items and subjects.

We next propose a fast SVD-based spectral method to estimate GoM models. Recall that Lemma 1 establishes the expressions for $\mathbf{\Pi}, \mathbf{\Theta}$ in (8) and (9). In practice, since $\mathbf{S}$ is not known, we propose to estimate it using a vertex-hunting technique called *sequential projection algorithm* (SPA; Araújo et al., 2001; Gillis and Vavasis, 2013). As stated in Lemma 1, Condition 1 induces a simplex geometry on the row vectors of $\mathbf{U}$ and the simplex vertices correspond to the pure subjects $\mathbf{S}$. To locate K vertices for any input matrix $\mathbf{U} \in \mathbb{R}^{N \times K}$ that has such a simplex structure, SPA first finds the subject with the maximum row norm in $\mathbf{U}$. That is, the first vertex index is

$$\hat{S}_1 = \operatorname{argmax}_{1 \leq i \leq N} \|\mathbf{U}_{i,:}\|_2.$$

Here and after we use $\|\mathbf{x}\|_2 = \sqrt{\mathbf{x}^\top \mathbf{x}}$ to denote the ℓ_2 norm of any vector $\mathbf{x}$. SPA then projects all the remaining row vectors $\{\mathbf{U}_{i,:} : i \neq \hat{S}_1\}$ onto the subspace that is orthogonal to $\mathbf{U}_{\hat{S}_1,:}$. Mathematically, denote $\mathbf{v}_1 := \mathbf{U}_{\hat{S}_1,:} / \|\mathbf{U}_{\hat{S}_1,:}\|_2$ as the scaled vector with unit norm, then the projected vectors are the rows of the matrix $\mathbf{U}(\mathbf{I}_K - \mathbf{v}_1 \mathbf{v}_1^\top)$, where $\mathbf{I}_K - \mathbf{v}_1 \mathbf{v}_1^\top$ is a $K \times K$ projection matrix. In the second step, SPA finds the second vertex index as the one that has the maximum norm among the projected row vectors,

$$\hat{S}_2 = \operatorname{argmax}_{i \neq \hat{S}_1} \|\mathbf{U}_{i,:}(\mathbf{I}_K - \mathbf{v}_1 \mathbf{v}_1^\top)\|_2.$$

The above procedure of first finding the row with the maximum projected norm and then projecting the remaining rows onto the orthogonal space is iteratively employed until finding all K vertex indices. Sequentially, for each $k = 1, \dots, K - 1$, define a unit-norm vector $\mathbf{v}_k = \mathbf{U}_{\hat{S}_k,:} / \|\mathbf{U}_{\hat{S}_k,:}\|_2$, then the $(k + 1)$ -th vertex index is estimated as

$$\hat{S}_{k+1} = \operatorname{argmax}_{i \notin \{\hat{S}_1, \dots, \hat{S}_k\}} \|\mathbf{U}_{i,:}(\mathbf{I}_K - \mathbf{v}_1 \mathbf{v}_1^\top) \cdots (\mathbf{I}_K - \mathbf{v}_k \mathbf{v}_k^\top)\|_2.$$

Here the projection matrices $(\mathbf{I}_K - \mathbf{v}_1 \mathbf{v}_1^\top), \dots, (\mathbf{I}_K - \mathbf{v}_k \mathbf{v}_k^\top)$ sequentially project the rows of $\mathbf{U}$ to the orthogonal spaces of those already found vertices $\mathbf{U}_{\hat{S}_1,:}, \dots, \mathbf{U}_{\hat{S}_k,:}$. This SPA procedure can be intuitively understood by visually inspecting the toy example in Figure 1. Since $\mathbf{U}_{1,:}, \dots, \mathbf{U}_{N,:}$ lie in a triangle in Figure 1, it is not hard to see that the vector with the largest norm should be one of the three vertices $\mathbf{U}_{S_1,:}, \mathbf{U}_{S_2,:}, \mathbf{U}_{S_3,:}$, say $\mathbf{U}_{S_3,:}$. Furthermore, after projecting the remaining $\mathbf{U}_{i,:}$ onto the space orthogonal to $\mathbf{U}_{S_3,:}$, the maximum norm of the projected vectors should correspond to $i = S_1$ or S_2 . This observation intuitively justifies that SPA can find the correct set of pure subjects given a simplex structure. With the estimated pure subjects $\hat{\mathbf{S}}, \mathbf{\Pi}$ and $\mathbf{\Theta}$ can be subsequently obtained via (8) and (9).

The above estimation procedure is based on the assumption that $\mathbf{R}_0$ is known. In prac-

tice, we only have access to the binary random data matrix $\mathbf{R}$ whose expectation is $\mathbf{R}_0$. Fortunately, it is known that for a large-dimensional random matrix with a low-rank expectation, the top- K SVD of the random matrix and the SVD of its expectation are close (e.g., see [Chen et al., 2021a](#)). This nontrivial theoretical result is our key insight and motivation to consider the top- K SVD of $\mathbf{R}$ as a surrogate for that of $\mathbf{R}_0$:

$$\mathbf{R} \approx \hat{\mathbf{U}} \hat{\mathbf{\Sigma}} \hat{\mathbf{V}}^\top, \quad (11)$$

where $\hat{\mathbf{\Sigma}}$ is a $K \times K$ diagonal matrix collecting the K largest singular values of $\mathbf{R}$, and $\hat{\mathbf{U}}_{N \times K}$, $\hat{\mathbf{V}}_{J \times K}$ collect the corresponding left and right singular vectors with $\hat{\mathbf{U}}^\top \hat{\mathbf{U}} = \hat{\mathbf{V}}^\top \hat{\mathbf{V}} = \mathbf{I}_K$. Specifically, [Chen et al. \(2021a\)](#) proved that the difference between $\mathbf{U}$ and $\hat{\mathbf{U}}$ up to a rotation is small when N and J are large with respect to K . Therefore, since Lemma 1 shows that the population row vectors $\mathbf{U}_{1,:}, \dots, \mathbf{U}_{N,:}$ form a simplex structure, the empirical row vectors $\{\hat{\mathbf{U}}_{i,:}\}_{i=1}^N$ are expected to form a noisy simplex cloud distributed around the population simplex. We call this noisy cloud the empirical simplex. For an illustration of the population and the empirical simplex, see Figure 2 in Section 5.

In order to recover the population simplex from the empirical one, we use a pruning step similar to that in [Mao et al. \(2021\)](#) to reduce noise. We summarize this pruning procedure in Algorithm 1. The high-level idea behind pruning is that if $\hat{\mathbf{U}}_{i,:}$ has a large norm but very few close neighbors, then it is likely to be outside of the population simplex and hence should be pruned (i.e., removed) before performing SPA to achieve higher accuracy in vertex hunting. More specifically, Algorithm 1 first calculates the norm of each row of $\hat{\mathbf{U}}$ (line 1-3) and identifies the vectors with norms in the upper q -quantile (line 4). The larger q is, the more such points are found. Then for each vector found, its average distance x_i to its r nearest neighbors is calculated (line 5-8). Finally, the subjects to be pruned are those whose

Algorithm 1 Prune

Input: Empirical top- K left singular matrix $\hat{\mathbf{U}}$, the number of nearest neighbors r , two quantiles $q, e \in (0, 1)$.

Output: Set of subject indices $\hat{\mathbf{P}}$ to be pruned (i.e., removed)

```
1: for  $i \in 1, \dots, N$  do
2:    $l_i = \|\hat{\mathbf{U}}_{i,:}\|_2$ 
3: end for
4:  $\hat{\mathbf{P}}_0 = \{i : l_i \geq \text{upper-}q \text{ quantile of } \{l_i : 1 \leq i \leq N\}\}$ 
5: for  $i \in \hat{\mathbf{P}}_0$  do
6:    $\mathbf{d}_i = \{\text{the distances from } \hat{\mathbf{U}}_i \text{ to its } r \text{ nearest neighbors}\}$ 
7:    $x_i = \text{average}(\mathbf{d}_i)$ 
8: end for
9:  $\hat{\mathbf{P}} = \{i : x_i \geq \text{upper-}e \text{ quantile of } \{x_i : i \in \hat{\mathbf{P}}_0\}\}$ 
```

x_i belong to the upper e -quantile of all the x_i 's (line 9). A larger value of e indicates that a larger proportion of the points will be pruned. According to our preliminary simulations, we observe that the estimation results are not very sensitive to these tuning parameters r, q, e . After pruning, we can use SPA to hunt for the K vertices of the pruned empirical simplex to obtain $\hat{\mathbf{S}}$, and then estimate $\mathbf{\Pi}$ and $\mathbf{\Theta}$.

Algorithm 2 summarizes our proposed method of estimating parameters $(\mathbf{\Pi}, \mathbf{\Theta})$ based on SPA with the pruning step. We first introduce some notation. The index of the element in $\mathbf{x}$ that has the maximum value is denoted by $\text{argmax}(\mathbf{x})$. For any matrix $\mathbf{A} = (a_{ij})$, denote by $\mathbf{A}_+ = (\max\{a_{ij}, 0\})$ the matrix that retains the nonnegative values of $\mathbf{A}$ and sets any negative values to zero. Denote by $\text{diag}(\mathbf{x})$ the diagonal matrix whose diagonals are the entries of the vector $\mathbf{x}$. If $\mathbf{S}_2$ is a subvector of $\mathbf{S}_1$, denote by $\mathbf{S}_1 \setminus \mathbf{S}_2$ the complement of vector $\mathbf{S}_2$ in vector $\mathbf{S}_1$. For a positive integer M , denote $[M] = \{1, \dots, M\}$. After obtaining the index vector of the pruned subjects $\hat{\mathbf{P}}$ (which is a subvector of $(1, 2, \dots, N)$) from Algorithm 1 (line 2), we use SPA on the pruned matrix $\hat{\mathbf{U}}_{[N] \setminus \hat{\mathbf{P}},:}$ to obtain the estimated pure subject index vector $\hat{\mathbf{S}}$ (line 3-8). Once this is achieved, $\mathbf{\Pi}$ and $\mathbf{\Theta}$ can be estimated by modifying

Algorithm 2 GoM Estimation by Sequential Projection Algorithm with Pruning

Input: Binary response matrix $\mathbf{R} \in \{0, 1\}^{N \times J}$, number of extreme profiles K , threshold parameter ϵ , pruning parameters r, q, e .

Output: Estimated $\hat{\mathbf{S}}, \hat{\mathbf{\Theta}}, \hat{\mathbf{\Pi}}$

- 1: Get the top K singular value decomposition of $\mathbf{R}$ as $\hat{\mathbf{U}}\hat{\mathbf{\Sigma}}\hat{\mathbf{V}}^\top$
 - 2: $\hat{\mathbf{P}} = \text{Prune}(\hat{\mathbf{U}}, r, q, e)$
 - 3: $\mathbf{Y} = \hat{\mathbf{U}}$
 - 4: **for** $k \in \{1, \dots, K\}$ **do**
 - 5: $\hat{S}_k = \text{argmax}(\{\|\mathbf{Y}_{i,:}\|_2 : i \in [N] \setminus \hat{\mathbf{P}}\})$
 - 6: $\mathbf{u} = \mathbf{Y}_{\hat{S}_k,:} / \|\mathbf{Y}_{\hat{S}_k,:}\|_2$
 - 7: $\mathbf{Y} = \mathbf{Y}(\mathbf{I}_K - \mathbf{u}\mathbf{u}^\top)$
 - 8: **end for**
 - 9: $\tilde{\mathbf{\Pi}} = \hat{\mathbf{U}}(\hat{\mathbf{U}}_{\hat{\mathbf{S}},:})^{-1}$
 - 10: $\hat{\mathbf{\Pi}} = \text{diag}(\tilde{\mathbf{\Pi}}_+ \mathbf{1}_K)^{-1} \tilde{\mathbf{\Pi}}_+$
 - 11: $\tilde{\mathbf{\Theta}} = \hat{\mathbf{V}}\hat{\mathbf{\Sigma}}\hat{\mathbf{U}}^\top \hat{\mathbf{\Pi}}(\hat{\mathbf{\Pi}}^\top \hat{\mathbf{\Pi}})^{-1}$
 - 12: Write $\hat{\mathbf{\Theta}}$ as $(\hat{\theta}_{j,k})$, where $\hat{\theta}_{j,k} = \begin{cases} \tilde{\theta}_{j,k} & \text{if } \epsilon \leq \tilde{\theta}_{j,k} \leq 1 - \epsilon \\ \epsilon & \text{if } \tilde{\theta}_{j,k} < \epsilon \\ 1 - \epsilon & \text{if } \tilde{\theta}_{j,k} > 1 - \epsilon \end{cases}$
 ($\epsilon \geq 0$ can be set to zero. In the numerical studies we choose $\epsilon = 0.001$ to be consistent and comparable with the default setting in the JML function in the R package `sirt`.)
-

(8) and (9) in Lemma 1. We first calculate

$$\tilde{\mathbf{\Pi}} = \hat{\mathbf{U}} \left(\hat{\mathbf{U}}_{\hat{\mathbf{S}},:} \right)^{-1} \quad (12)$$

based on (8). The $\tilde{\mathbf{\Pi}}$ obtained above does not necessarily fall into the parameter domain for $\mathbf{\Pi}$. Therefore, we first truncate all the entries of $\tilde{\mathbf{\Pi}}$ to be nonnegative, and then re-normalize each row to sum to one (line 10). Based on $\hat{\mathbf{\Pi}}$, we can also estimate $\mathbf{\Theta}$ by

$$\tilde{\mathbf{\Theta}} = \hat{\mathbf{V}}\hat{\mathbf{\Sigma}}\hat{\mathbf{U}}^\top \hat{\mathbf{\Pi}}(\hat{\mathbf{\Pi}}^\top \hat{\mathbf{\Pi}})^{-1} \quad (13)$$

according to (9). Lastly, we truncate $\tilde{\mathbf{\Theta}}$ to be between $[\epsilon, 1 - \epsilon]$ and obtain the final estimator $\hat{\mathbf{\Theta}}$ (line 12). When $\epsilon = 0$, this truncation ensures that entries of $\hat{\mathbf{\Theta}}$ lie in the parameter domain $[0, 1]$. In the numerical studies, we choose $\epsilon = 0.001$ to be consistent and comparable

with the default setting in the JML function in the R package `sirt`.

Remark 1. *There are two possible ways to estimate Θ according to (9). The first one is defining $\hat{\Theta}$ as the truncated version of*

$$\tilde{\Theta} = \hat{\mathbf{V}}\hat{\Sigma}\hat{\mathbf{U}}_{\hat{\mathbf{S}},:}^{\top},$$

which only uses the information of $\hat{\mathbf{U}}$ corresponding to the K pure subjects indexed by $\hat{\mathbf{S}}$. Another approach is estimating Θ via (13), which uses information from all of the N subjects. The latter approach is expected to give more stable estimates. Our preliminary simulations also justify that using (13) indeed gives higher estimation accuracy, so we choose to estimate Θ that way.

5 Simulation Studies

5.1 Evaluating the Proposed Method

We carry out simulation studies to evaluate the accuracy and computational efficiency of our new method. We consider $K \in \{3, 8\}$, $N \in \{200, 1000, 2000, 3000, 4000, 5000\}$, and $J = N/5$, which correspond to large-scale and high-dimensional data scenarios. Such simulation regimes share a similar spirit with those in [Zhang et al. \(2020\)](#), which proposed and evaluated an SVD-based approach for item factor analysis. In each simulation setting, we generate 100 independent replicates. Within each setting and each replication, the rows of $\mathbf{\Pi}$ are independently simulated from the Dirichlet(α) distribution with α equal to the K -dimensional all-one vector, and the first K rows of $\mathbf{\Pi}$ are set to the identity matrix $\mathbf{I}_K$ in order to satisfy the pure-subject Condition 1. The entries in Θ are independently simulated from the uniform distribution on $[0, 1]$. Our preliminary simulations suggest setting

the tuning parameters in the pruning Algorithm 1 to $r = 10$, $q = 0.4$, $e = 0.2$ with 8% subjects removed before SPA, so that the pruned empirical simplex is adequately close to the population simplex. We use the above setting throughout all the simulations and the real data analysis unless otherwise specified. It turns out the performance of our method is robust to these tuning parameter values and not much tuning is needed in practice.

To illustrate the simplex geometry of the population and empirical left singular matrices $\mathbf{U}$ (from the SVD of $\mathbf{R}_0$) and $\hat{\mathbf{U}}$ (from the top- K SVD of $\mathbf{R}$), we plot $\mathbf{U}_{i,:}$ and $\hat{\mathbf{U}}_{i,:}$ with $N = 2000$ and $K = 3$ in Figure 2. All vectors are projected to two dimensions for better visualization of the simplex (i.e., triangle) structure. In Figure 2, the red-shaded area corresponds to the population simplex, the green crosses are the removed subjects selected by the pruning Algorithm 1, and the blue dots form the empirical simplex obtained after pruning. As we can see, the empirical row vectors $\hat{\mathbf{U}}_{i,:}$ approximately form a simplex cloud around the population simplex. After pruning out the noisy vectors, the resulting empirical simplex is close to the population one. This fact not only illustrates the effectiveness of the pruning procedure in Algorithm 1, but also confirms the usefulness of our notion of expectation identifiability by showing the close proximity of $\mathbf{U}$ and $\hat{\mathbf{U}}$. The resemblance between the scatter plots of the rows of $\mathbf{U}$ and $\hat{\mathbf{U}}$ implies that the simplex geometry of $\mathbf{U}$ holds approximately for $\hat{\mathbf{U}}$, and hence justifies that SPA can be applied to $\hat{\mathbf{U}}$ to estimate $\mathbf{\Pi}$ and $\mathbf{\Theta}$.

We compare the performance of our proposed method with the JML algorithm (Joint Maximum Likelihood) in the R package `sirt`, because the latter is currently considered as the most efficient estimation method for GoM models. We follow the default settings of JML with the maximum iteration number set as 600, the global parameter convergence criterion as 0.001, the maximum change in relative deviance as 0.001, and the minimum value of π_{ik} and θ_{jk} as 0.001. We measure the parameter estimation error by the mean absolute error (MAE). That is, the error between the estimated $\hat{\mathbf{\Pi}}$ and the ground truth $\mathbf{\Pi}$ for each

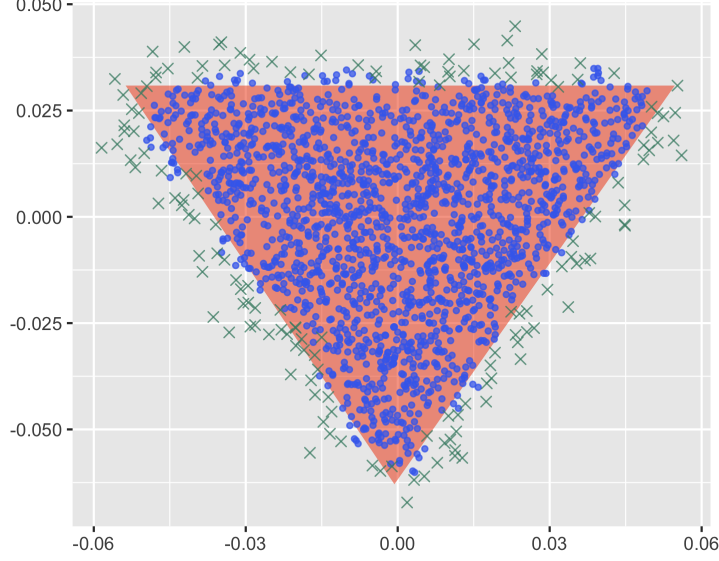


Figure 2: Row vectors of $\mathbf{U}$ and $\hat{\mathbf{U}}$ projected to $\mathbb{R}^2$ in the simulation setting with $N = 2000$ and $K = 3$. The red-shaded area is the population simplex, the green crosses are the removed subjects in pruning, and the blue dots form the empirical simplex retained after pruning.

replicate is quantified by the mean absolute bias:

$$l(\mathbf{\Pi}, \hat{\mathbf{\Pi}}) := \frac{1}{NK} \sum_{i=1}^N \sum_{k=1}^K |\pi_{ik} - \hat{\pi}_{ik}|.$$

Similarly,

$$l(\mathbf{\Theta}, \hat{\mathbf{\Theta}}) := \frac{1}{JK} \sum_{j=1}^J \sum_{k=1}^K |\theta_{jk} - \hat{\theta}_{jk}|.$$

We present the comparisons between our spectral method and JML in terms of computation time and estimation error in Table 1 and Figures 3, 4, 5. As shown in Figure 3, the computation time for both methods increases as the sample size N and the number of items J increase. Notably, JML takes significantly more time than our proposed method, especially when N and J are large. For example, when $N = 5000$, $J = 1000$, $K = 8$, it takes about 3 hours for JML to reach the maximum iteration number for each replication, while it takes less than 40 seconds on average for our proposed method. Table 1 further records the

mean computational time in seconds across replications for JML and the proposed method. Moreover, we observe that JML is not able to converge in almost all replications when $K = 8$, including when the sample size is as small as $N = 200$. In summary, when the number of extreme latent profiles K is large enough, JML not only takes a long time to run but also is unable to reach convergence given the default convergence criterion.

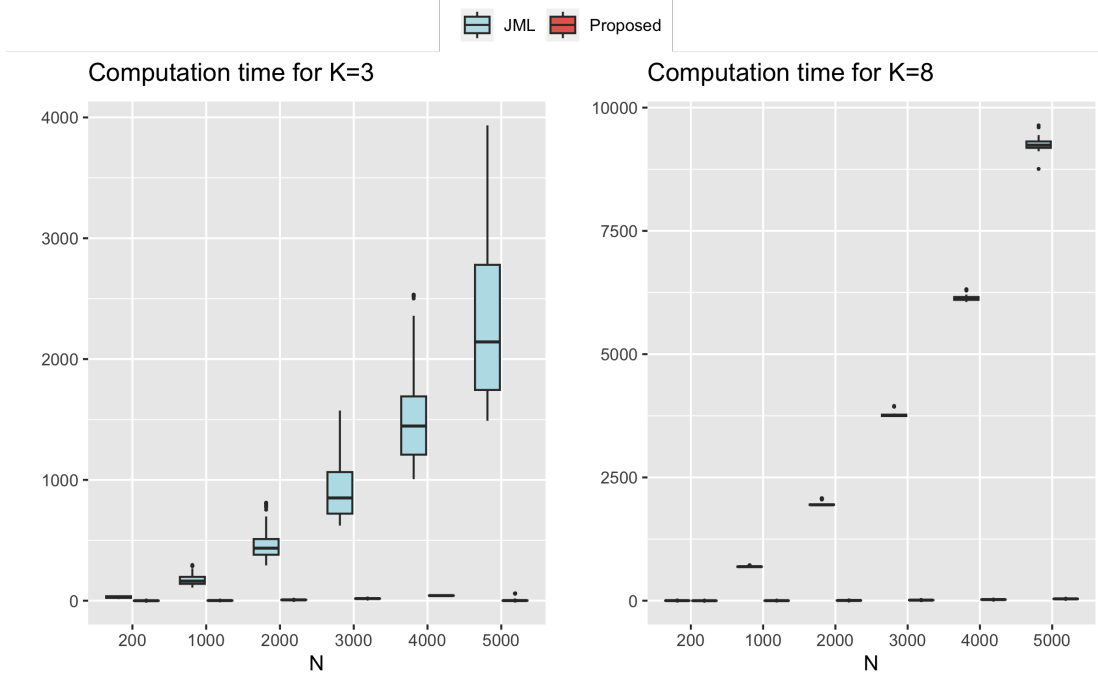


Figure 3: Computation time for $K = 3$ (left) and $K = 8$ (right) in simulations. The boxplots represent the run time in seconds versus the sample size N .

K	Method	$N = 200$	$N = 1000$	$N = 2000$	$N = 3000$	$N = 4000$	$N = 5000$
$K = 3$	JML	30.3	170.9	473.9	923.4	1528.4	2363.9
	Proposed	7.5e-2	1.5	6.8	17.5	42.3	10.6
$K = 8$	JML	1.7	691.5	1957.5	3761.7	6134.3	9251.7
	Proposed	6.4e-2	1.3	5.0	11.6	23.0	37.2

Table 1: Table of average computational time in seconds across replications for JML and the proposed spectral method for $K = 3$ and $K = 8$.

The huge computational advantage of the proposed method does not come at the cost of degraded estimation accuracy. Figures 4 and 5 show that the estimation error decreases as

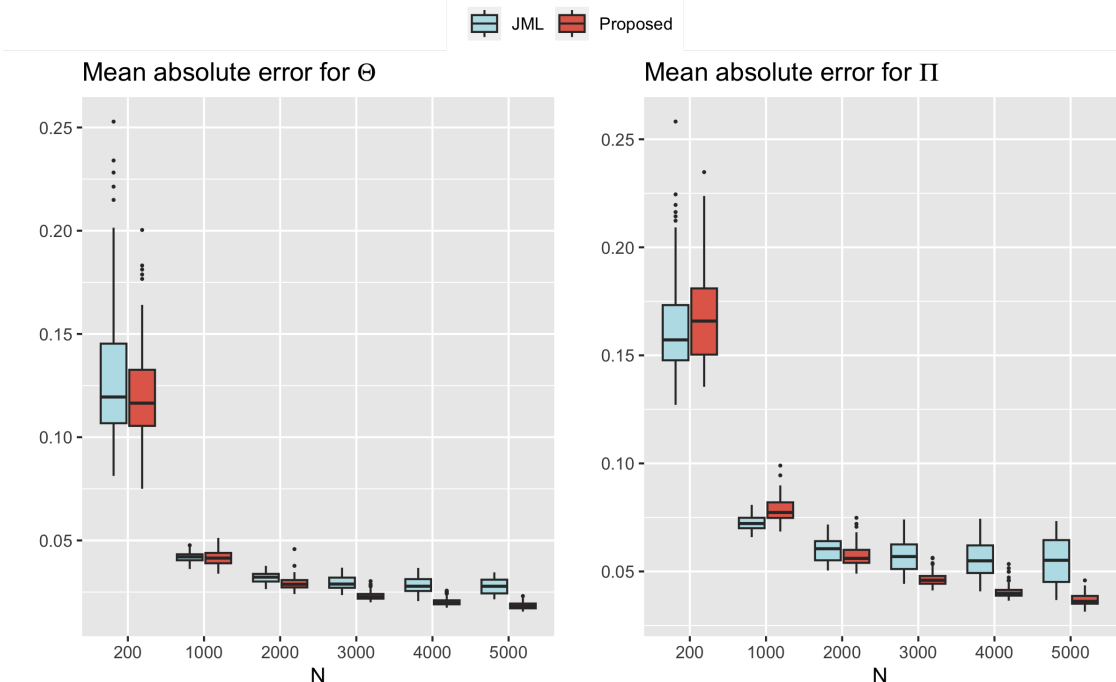


Figure 4: Simulation results of estimation error for $K = 3$. The boxplots represent the mean absolute error for Θ (left) and Π (right) versus the sample size N .

the sample size N and the number of items J increase for both methods. When the sample size is large enough ($N \geq 2000$) for $K = 3$, our proposed method gives more accurate estimation on average compared to JML. For $K = 8$, our proposed method yields higher estimation accuracy for Θ for all sample sizes on average. When $N \geq 2000$, the estimation accuracy of Π is slightly worse but comparable to JML. We point out that due to their non-iterative nature, SVD or eigendecomposition-based methods typically tend to give worse estimation compared to iterative methods that aim to find the MLE; for example, see the comparison of an SVD-based method and a joint MLE method for item factor analysis in [Zhang et al. \(2020\)](#). However, it turns out that given a fixed computational resource (i.e., the default maximum iteration number in the `sirt` package), the iterative method JML can give worse estimation accuracy than our spectral method.

To summarize, the above simulation results demonstrate the superior efficiency and accuracy of our proposed method to estimate GoM models. Our method provides comparable

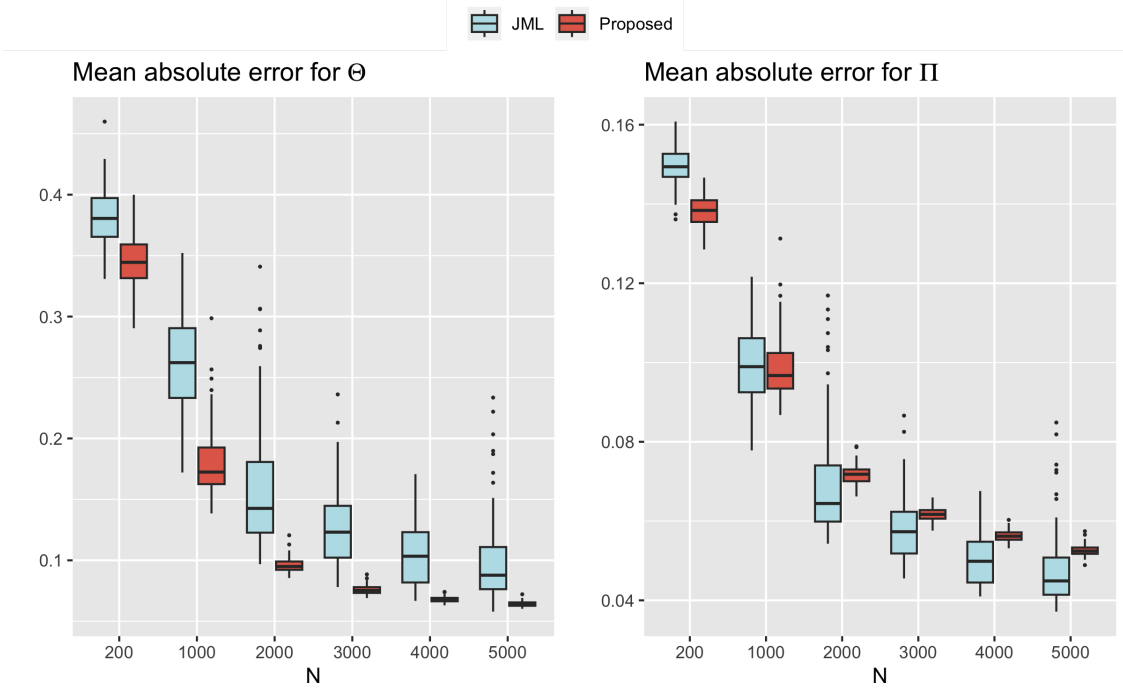


Figure 5: Simulation results of estimation error for $K = 8$. The boxplots represent the mean absolute error for Θ (left) and Π (right) versus the sample size N .

or even better estimation results compared with JML, and is much more scalable to large datasets with many subjects and many items.

5.2 Verifying Identifiability

We also conduct another simulation study to verify the identifiability results. We consider three different cases with $K = 3$, $N \in \{200, 1000, 2000, 3000\}$, and $J = N/5$. In Case 1, we set the ground truth Θ by vertically concatenating copies of the identifiable 4×3 Θ -matrix in Example 1, and the generation mechanism for Π remains the same as in Section 5.1. In Case 2, we use the same Θ as in Case 1; while for Π , after generating rows of $\Pi' = (\pi'_{ik})$ from $\text{Dirichlet}(\mathbf{1})$, we truncate π'_{ik} to be no less than $1/3$ and then re-normalize each row of it; after this operation the minimum entry in the resulting Π is 0.2. Such a generated $\Pi = (\pi_{ik})$ does not satisfy the pure-subject Condition 1. In Case 3, we generate Π using

the same mechanism as in Section 5.1, but generate Θ whose rows are replicates of the vector $(0.8, 0.5, 0.2)$ so that $\text{rank}(\Theta) = 1$. Case 1 falls into part (a) in Theorem 2 and is identifiable, whereas Cases 2 and 3 correspond to part (c) and are not identifiable. Figure 6 shows that the estimation errors in Cases 2 and 3 are significantly larger than those in Case 1. These results empirically verify our identifiability conclusions in Theorem 2.

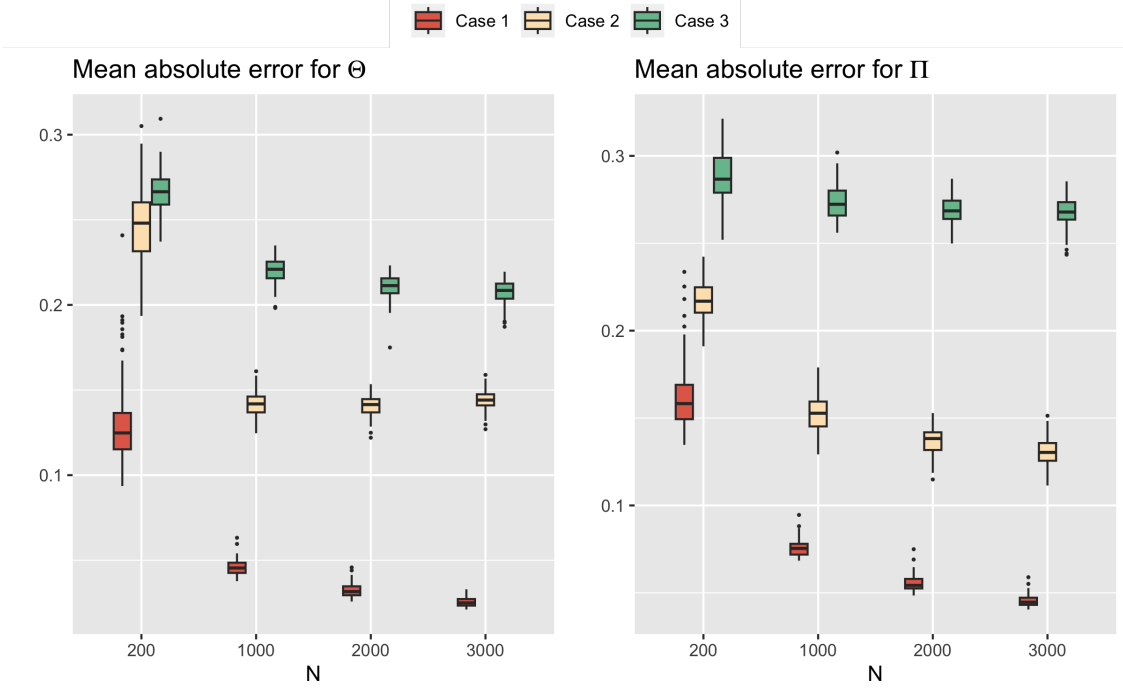


Figure 6: A simulation study verifying identifiability. Estimation errors for three different cases; see the concrete settings of Cases 1, 2, and 3 in the main text. The box plots represent the mean absolute error for Θ (left) and Π (right) versus the sample size N .

6 Real Data Example

We illustrate our proposed method by applying it to a real-world personality test dataset, the Woodworth Psychoneurotic Inventory (WPI) dataset. WPI was designed by the United States Army during World War I to identify soldiers at risk for shell shock and is often credited as the first personality test. The WPI dataset can be downloaded from the Open

Psychometrics Project website: http://openpsychometrics.org/_rawdata/. The dataset consists of binary yes/no responses to $J = 116$ items from 6019 subjects. We remove subjects with missing responses and only keep the subjects who are at least ten years old. This screening process leaves us with $N = 3842$ subjects.

We apply both the JML method and our new spectral method to the WPI dataset to compare computation time. Figure 7 shows the computation time in seconds versus the number of extreme profiles K for the two methods. For $K \geq 4$, the number of iterations for JML reaches the default maximum iteration number in the `sirt` package and does not converge. Similar to the simulations, our spectral method takes significantly less computation time compared to JML. This observation again confirms that the proposed method is scalable to real-world datasets with a large sample size and a relatively large number of items.

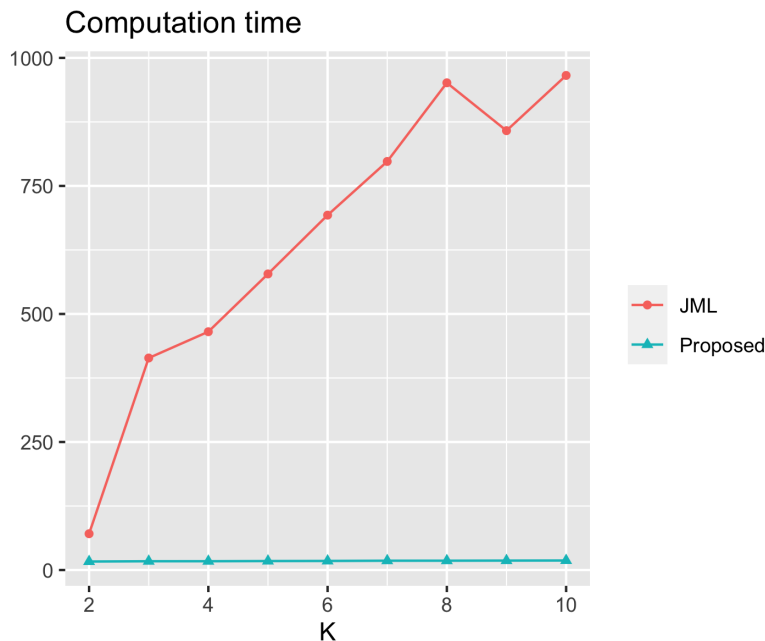


Figure 7: Computation time for the WPI dataset. The lines indicate the run time in seconds versus K for JML and our spectral method. Note that for $K \geq 4$, the number of iterations in JML reaches the default maximum iteration number.

Choosing the number of extreme latent profiles K is a nontrivial problem for GoM models in practice. Available model selection techniques include the Akaike information crite-

rion (AIC; Akaike, 1998) and the Bayesian information criterion (BIC; Schwarz, 1978) for likelihood-based methods, and the deviance information criteria (DIC; Spiegelhalter et al., 2002) for Bayesian MCMC methods. For factor analysis and principal component analysis, parallel analysis (Horn, 1965; Dobriban and Owen, 2019) is one popular eigenvalue-based method to select the latent dimension. For the WPI dataset, when choosing $K = 3$, we observe that the estimated $\hat{\Theta}$ matrix has three well-separated column vectors, which imply a meaningful interpretation of the extreme profiles. When increasing K to 4, the columns of the estimated $\hat{\Theta}$ are no longer that well-separated and interpretable. Moreover, choosing $K = 3$ produces the least reconstruction error compared to $K = 2$ or 4. That is, $K = 3$ leads to the smallest mean absolute error between $\hat{\Pi}\hat{\Theta}^\top$ and the observed data matrix $\mathbf{R}$. Since the goal of the current data analysis is mainly to illustrate the proposed spectral method, we next present and discuss the estimation results for $K = 3$. The important problem of how to select K in GoM models in a principled manner is left as a future direction.

We next take a closer look at the estimation result given by the proposed method for $K = 3$. In order to interpret each extreme latent profile, we present the heatmap of part of the estimated item parameter matrix $\hat{\Theta}$ in Figure 8. Since the number of items $J = 116$ is large, we only display a subset of the items for better visualization. More specifically, 30 out of 116 items are chosen in Figure 8 based on the criterion that the chosen items have the largest variability in $(\hat{\theta}_{j,1}, \hat{\theta}_{j,2}, \hat{\theta}_{j,3})$. Based on Figure 8, we interpret profile 1 as people who are socially passive, profile 3 as physically unhealthy, and profile 2 as otherwise healthy. The estimated item parameters for profile 1 show that people belonging to this profile tend to be shy, worried, and easy to get tired of people or things. People belonging to profile 3 have higher probabilities of fainting, dyspepsia, asthma, and hay fever.

Figure 9 shows the ternary diagram of the estimated membership scores $\hat{\Pi}$ made with the R package `ggtern`. The WPI dataset comes with the age information of each subject, and

Were you shy with other boys [or girls]?	0.893	0.497	0.221
Do you worry too much about little things?	0.848	0.545	0.207
Do you get tired of people quickly?	0.822	0.459	0.1
Is it easy to make you laugh?	0.755	0.557	0.342
Did you ever have a strong desire to run away from home?	0.97	0.754	0.183
Do you get tired of work?	0.999	0.699	0.189
Have you ever had fits of dizziness?	0.999	0.568	0.162
Do your interests change frequently?	0.999	0.248	0.382
Did the other children let you play with them?	0.972	0.915	0.354
Can you do the little chores of the day without worrying over them?	0.935	0.842	0.345
Do you like outdoor life?	0.999	0.892	0.518
Do you think you worry too much when you have an unfinished job on your hands?	0.752	0.963	0.211
As a child did you like to play alone better than to play with other children?	0.779	0.773	0.228
Do you have the sensation of falling when going to sleep?	0.999	0.927	0.089
Did the teachers in school generally treat you right?	0.675	0.323	0.361
Do you get tired of amusements quickly?	0.75	0.17	0.318
Is it easy to get you cross or grouchy?	0.473	0.084	0.249
Do you find your way about easily?	0.465	0.763	0.999
Can you stand the sight of blood?	0.415	0.791	0.999
Have you ever fainted away?	0.317	0.597	0.999
Can you stand pain quietly?	0.45	0.974	0.938
Have your employers generally treated you right?	0.423	0.991	0.914
Can you sit still without fidgeting?	0.775	0.778	0.954
Can you stand disgusting smells?	0.751	0.669	0.962
Do you get used to new places quickly?	0.612	0.77	0.96
Did you ever have dyspepsia [indigestion]?	0.484	0.396	0.999
Did you ever have asthma or hay fever [allergies]?	0.498	0.37	0.999
Do ideas run through your head so that you cannot sleep?	0.446	0.353	0.999
Have you a good appetite?	0.358	0.406	0.825
Did you ever have the habit of biting your finger nails?	0.221	0.384	0.912
	profile 1	profile 2	profile 3

Figure 8: Heatmap of $\hat{\Theta}$ of a subset of 30 WPI items. The values are the estimated probability of responding “yes” for each item given each extreme profile.

we color-code the subjects in Figure 9 according to their ages. The darker color represents older people while the lighter color represents younger people. Each dot represents a subject, and the location of the dot in the equilateral triangle depicts the three membership scores of this subject. Specifically, dots with a large membership score on a profile are closer to the vertex of this profile. One can see that the pure-subject Condition 1 is satisfied here since there are dots located almost exactly at each of the three vertices in Figure 9.

This figure also reveals that darker dots are more gathered around the vertex of profile 3, which means that older people are more likely to belong to the extreme profile identified as physically unhealthy. Correspondingly, younger people are closer to profile 1 or 2. Recalling our earlier interpretation of the extreme profiles from Figure 8, these results are intuitively meaningful since older people tend to be less healthy compared with younger people. It is worth emphasizing that the age information is *not used* in our estimation of the GoM model, but our method is able to generate interpretable results with respect to age.

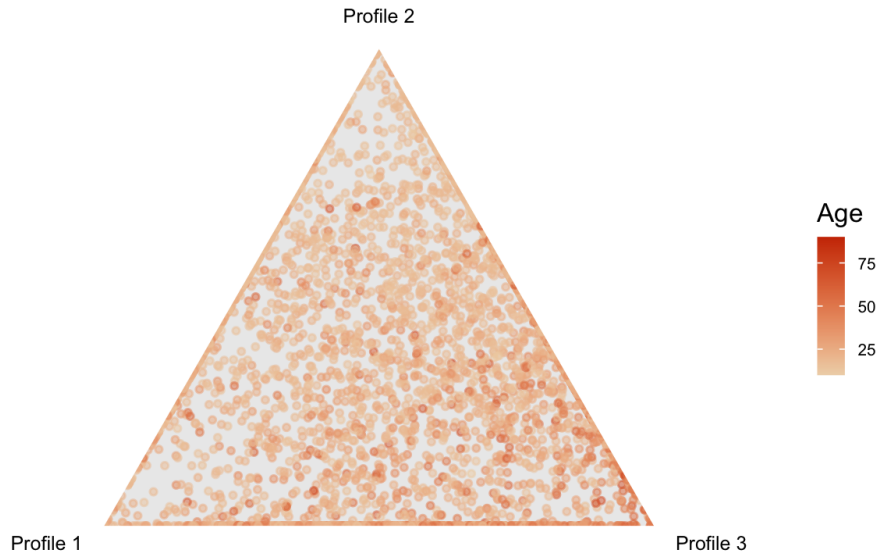


Figure 9: The barycentric plot of the estimated membership scores $\hat{\Pi}$ for WPI data, color-coded with the age covariate.

7 Discussion

In this paper, we have adopted a spectral approach to GoM analysis of multivariate binary responses. Under the notion of expectation identifiability, we have proposed sufficient conditions that are close to being necessary for GoM models to be identifiable. For estimation, we

have proposed an efficient SVD-based spectral algorithm to estimate the subject-level and population-level parameters in the GoM model. Our spectral method has a huge computational advantage over Bayesian or likelihood-based methods and is scalable to large-scale and high-dimensional data. Simulation results demonstrate the superior efficiency and accuracy of our method, and also empirically corroborate our identifiability conclusions. We hope this work provides a useful tool for psychometric researchers and practitioners by making GoM analysis less computationally daunting and statistically mysterious.

There are several future research directions worth exploring. First, building on the identifiability results, it would be interesting to derive statistical error bounds to show how the parameter estimation error explicitly depends on N , J , and K , similar in spirit to [Zhang et al. \(2020\)](#) for item factor analysis. Second, this work has focused on binary responses, while in practice it is also of interest to perform GoM analysis of polytomous responses, such as Likert-scale responses in psychological questionnaires (see, e.g. [Gu et al., 2023](#)). It would be desirable to extend our method to such multivariate polytomous data. Under a GoM model for polytomous data, a similar low-rank structure as (5) should still exist for each category of the responses. Since our method is built upon the low-rank structure and the pure subject condition, we conjecture that exploiting such structures in polytomous data could still lead to nice spectral algorithms. Third, it is worth considering developing a model that directly incorporates additional covariates into the GoM analysis. Our current method does not use additional covariates, such as the age information in the WPI dataset. Proposing a covariate-assisted GoM model and a corresponding spectral method may be methodologically interesting and practically useful.

Appendix A Proofs of the Identifiability Results

Proof of Proposition 1. If we take the rows that correspond to $\mathbf{S}$ of both sides in the SVD (6) and use the fact that $\mathbf{\Pi}_{\mathbf{S},:} = \mathbf{I}_K$, then

$$\mathbf{U}_{\mathbf{S},:}\mathbf{\Sigma}\mathbf{V}^\top = [\mathbf{R}_0]_{\mathbf{S},:} = \mathbf{\Pi}_{\mathbf{S},:}\mathbf{\Theta}^\top = \mathbf{\Theta}^\top. \quad (14)$$

This gives an expression of $\mathbf{\Theta}$

$$\mathbf{\Theta} = \mathbf{V}\mathbf{\Sigma}\mathbf{U}_{\mathbf{S},:}^\top. \quad (15)$$

Further note that

$$\mathbf{U} = \mathbf{R}_0\mathbf{V}\mathbf{\Sigma}^{-1} = \mathbf{\Pi}\mathbf{\Theta}^\top\mathbf{V}\mathbf{\Sigma}^{-1}. \quad (16)$$

If we plug (14) into (16) and note that $\mathbf{V}$ have orthogonal columns, we have

$$\mathbf{U} = \mathbf{\Pi}\mathbf{U}_{\mathbf{S},:}\mathbf{\Sigma}\mathbf{V}^\top\mathbf{V}\mathbf{\Sigma}^{-1} = \mathbf{\Pi}\mathbf{U}_{\mathbf{S},:}. \quad (17)$$

Equation (17) also tells us that $\mathbf{U}_{\mathbf{S},:}$ must be full rank since both $\mathbf{U}$ and $\mathbf{\Pi}$ have rank K .

Therefore, we have an expression of $\mathbf{\Pi}$:

$$\mathbf{\Pi} = \mathbf{U}(\mathbf{U}_{\mathbf{S},:})^{-1}.$$

On the other hand, based on the singular value decomposition $\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top = \mathbf{\Pi}\mathbf{\Theta}^\top$, we can left multiply $(\mathbf{\Pi}^\top\mathbf{\Pi})^{-1}\mathbf{\Pi}^\top$ with both hand sides to obtain

$$\begin{aligned} (\mathbf{\Pi}^\top\mathbf{\Pi})^{-1}\mathbf{\Pi}^\top\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top &= \mathbf{\Theta}^\top \\ \Rightarrow \mathbf{\Theta} &= \mathbf{V}\mathbf{\Sigma}\mathbf{U}^\top\mathbf{\Pi}(\mathbf{\Pi}^\top\mathbf{\Pi})^{-1}. \end{aligned} \quad (18)$$

We next show that (15) and (18) are equivalent. Since $\mathbf{U}^\top \mathbf{U} = \mathbf{I}_K$, (17) leads to

$$\mathbf{U}_{\mathbf{s},:}^\top \mathbf{\Pi}^\top \mathbf{\Pi} \mathbf{U}_{\mathbf{s},:} = \mathbf{I}_K,$$

which yields

$$(\mathbf{\Pi}^\top \mathbf{\Pi})^{-1} = \mathbf{U}_{\mathbf{s},:} \mathbf{U}_{\mathbf{s},:}^\top.$$

Plugging this equation into (18), we have

$$\begin{aligned} \mathbf{\Theta} &= \mathbf{V} \mathbf{\Sigma} \mathbf{U}^\top \mathbf{\Pi} \mathbf{U}_{\mathbf{s},:} \mathbf{U}_{\mathbf{s},:}^\top \\ &= \mathbf{V} \mathbf{\Sigma} \mathbf{U}^\top \mathbf{U} (\mathbf{U}_{\mathbf{s},:})^{-1} \mathbf{U}_{\mathbf{s},:} \mathbf{U}_{\mathbf{s},:}^\top \\ &= \mathbf{V} \mathbf{\Sigma} \mathbf{U}_{\mathbf{s},:}^\top \end{aligned}$$

This shows the equivalence of (15) and (18) and completes the proof of the proposition. $\square$

Proof of Theorem 1. Without loss of generality, assume that the first extreme latent profile does not have a pure subject. Then $\pi_{i1} \leq 1 - \delta$, $\forall i = 1, \dots, N$ for some $\delta > 0$. For each $0 < \epsilon < \delta$, define a $K \times K$ matrix

$$\mathbf{M}_\epsilon = \begin{bmatrix} 1 + (K-1)\epsilon^2 & -\epsilon^2 \mathbf{1}_{K-1}^\top \\ \mathbf{0}_{K-1} & \epsilon \mathbf{1}_{K-1} \mathbf{1}_{K-1}^\top + (1 - (K-1)\epsilon) \mathbf{I}_{K-1} \end{bmatrix}.$$

We will show that $\tilde{\mathbf{\Pi}}_\epsilon = \mathbf{\Pi} \mathbf{M}_\epsilon$ and $\tilde{\mathbf{\Theta}}_\epsilon = \mathbf{\Theta} \mathbf{M}_\epsilon^{-1}$ form a valid parameter set. That is, each element of $\tilde{\mathbf{\Pi}}_\epsilon$ and $\tilde{\mathbf{\Theta}}_\epsilon$ lies in $[0, 1]$ and the rows of $\tilde{\mathbf{\Pi}}_\epsilon$ sum to one. Since $\mathbf{M}_0 = \mathbf{I}_K$ and by the continuity of matrix determinant, $\mathbf{M}_\epsilon$ is full rank when ϵ is small enough. Also notice that $\mathbf{M}_\epsilon \mathbf{1}_K = \mathbf{1}_K$. Therefore, $\tilde{\mathbf{\Pi}}_\epsilon \mathbf{1}_K = \mathbf{\Pi} \mathbf{M}_\epsilon \mathbf{1}_K = \mathbf{\Pi} \mathbf{1}_K = \mathbf{1}_N$. For each $i = 1, \dots, N$, $\tilde{\pi}_{i1} = \pi_{i1}(1 + (K-1)\epsilon^2) \geq 0$. For any fixed $k = 2, \dots, K$, $(\mathbf{M}_\epsilon)_{kk} = 1 - (K-2)\epsilon$ and $(\mathbf{M}_\epsilon)_{mk} = \epsilon$ for $m \neq k$. Thus when $\epsilon \leq 1/(K-1)$, we have $(\mathbf{M}_\epsilon)_{mk} \geq \epsilon$ for any $m = 1, \dots, K$.

Therefore, the following inequalities hold for each $i = 1, \dots, N$ and $k = 2, \dots, K$:

$$\begin{aligned}\tilde{\pi}_{ik} &= -\epsilon^2 \pi_{i1} + \sum_{m=2}^K \pi_{im} (\mathbf{M}_\epsilon)_{mk} \geq -\epsilon^2 \pi_{i1} + \epsilon \sum_{m=2}^K \pi_{im} \\ &\geq -\epsilon^2(1 - \delta) + \epsilon(1 - \pi_{i1}) \geq -\epsilon^2(1 - \delta) + \epsilon\delta \geq \epsilon\delta^2 > 0.\end{aligned}$$

Here we also used $\sum_{k=1}^K \pi_{ik} = 1$, $\pi_{i1} \leq 1 - \delta$, and $\epsilon < \delta$.

Further notice that

$$\mathbf{M}_\epsilon - \mathbf{I}_K = \begin{bmatrix} \epsilon^2(K-1) & -\epsilon^2 \mathbf{1}_{K-1}^\top \\ \mathbf{0}_{K-1} & \epsilon \mathbf{1}_{K-1} \mathbf{1}_{K-1}^\top - \epsilon(K-1) \mathbf{I}_{K-1} \end{bmatrix},$$

which leads to $\|\mathbf{M}_\epsilon - \mathbf{I}_K\|_F \xrightarrow{\epsilon \rightarrow 0} 0$. Here $\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2}$ is the Frobenius norm of any matrix $\mathbf{A} = (a_{ij}) \in \mathbb{R}^{m \times n}$. By the continuity of matrix inverse and Frobenius norm,

$$\|\mathbf{M}_\epsilon^{-1} - \mathbf{I}_K\|_F \xrightarrow{\epsilon \rightarrow 0} 0.$$

Therefore,

$$\|\tilde{\Theta} - \Theta\|_F = \|\Theta(\mathbf{I}_K - \mathbf{M}_\epsilon^{-1})\|_F \leq \|\Theta\|_2 \|\mathbf{I}_K - \mathbf{M}_\epsilon^{-1}\|_F \xrightarrow{\epsilon \rightarrow 0} 0.$$

Since all the elements of Θ are strictly in $(0, 1)$, the elements of $\tilde{\Theta}$ must be in $[0, 1]$ when ϵ is small enough. Also note that $\mathbf{M}_\epsilon$ is not a permutation matrix when $\epsilon > 0$, thus the GoM model is not identifiable up to a permutation. This completes the proof of the theorem. $\square$

Proof of Theorem 2. Suppose $\text{rank}(\Theta) = r \leq K$. Now, consider SVD $\mathbf{R}_0 = \mathbf{U}\Sigma\mathbf{V}^\top$ with $\mathbf{U} \in \mathbb{R}^{N \times r}$, $\mathbf{V} \in \mathbb{R}^{J \times r}$, $\Sigma \in \mathbb{R}^{r \times r}$. For simplicity, we continue to use the same notations $\mathbf{U}, \Sigma, \mathbf{V}$ here even though the matrix dimensions have changed. Without loss of generality, we can reorder the subjects and memberships so that $\Pi_{1:K,:} = \mathbf{I}_K$ from Assumption 1.

According Proposition 1,

$$\mathbf{U} = \mathbf{\Pi} \mathbf{U}_{1:K,:}. \quad (19)$$

Since $\text{rank}(\mathbf{\Pi}) = K, \text{rank}(\mathbf{U}) = r$, we must have $\text{rank}(\mathbf{U}_{1:K,:}) = \text{rank}(\mathbf{U}) = r$.

Suppose another set of parameters $(\tilde{\mathbf{\Pi}}, \tilde{\mathbf{\Theta}})$ yields the same $\mathbf{R}_0$ and we denote its corresponding pure subject index vector as $\tilde{\mathbf{S}}$ so that $\tilde{\mathbf{\Pi}}_{\tilde{\mathbf{S}},:} = \mathbf{I}_K$. Similarly, we have

$$\mathbf{U} = \tilde{\mathbf{\Pi}} \mathbf{U}_{\tilde{\mathbf{S}},:}. \quad (20)$$

Taking the $\tilde{\mathbf{S}}$ rows of both sides of (19) and the first K rows of both sides of (20) yields

$$\mathbf{\Pi}_{\tilde{\mathbf{S}},:} \mathbf{U}_{1:K,:} = \mathbf{U}_{\tilde{\mathbf{S}},:}, \quad \mathbf{U}_{1:K,:} = \tilde{\mathbf{\Pi}}_{1:K,:} \mathbf{U}_{\tilde{\mathbf{S}},:}.$$

The above equation shows that $\mathbf{U}_{\tilde{\mathbf{S}},:}$ is in the convex hull created by the rows of $\mathbf{U}_{1:K,:}$, and $\mathbf{U}_{1:K,:}$ is in the convex hull created by the rows of $\mathbf{U}_{\tilde{\mathbf{S}},:}$. Therefore, there must exist a permutation matrix $\mathbf{P}$ such that $\mathbf{U}_{\tilde{\mathbf{S}},:} = \mathbf{P} \mathbf{U}_{1:K,:}$. Combining this fact with (19) and (20) leads to

$$(\mathbf{\Pi} - \tilde{\mathbf{\Pi}} \mathbf{P}) \mathbf{U}_{1:K,:} = 0. \quad (21)$$

Proof of part (a). For part (a), $r = K$ and $\mathbf{U}_{1:K,:}$ is full rank according to (19). In this case, (21) directly leads to $\mathbf{\Pi} = \tilde{\mathbf{\Pi}} \mathbf{P}$ and thus $\tilde{\mathbf{\Theta}} = \mathbf{\Theta} \mathbf{P}^\top$.

Now generally consider $r < K$. By permuting the rows and columns of $\mathbf{\Theta}$, we can write

$$\mathbf{\Theta} = \begin{bmatrix} \mathbf{C} & \mathbf{C} \mathbf{W}_1 \\ \mathbf{W}_2^\top \mathbf{C} & \mathbf{W}_2^\top \mathbf{C} \mathbf{W}_1 \end{bmatrix}, \quad (22)$$

where $\mathbf{C} \in \mathbb{R}^{r \times r}$ is full rank, $\mathbf{W}_1 \in \mathbb{R}^{r \times (K-r)}$ and $\mathbf{W}_2 \in \mathbb{R}^{r \times (J-r)}$. Now comparing the block

columns of (22) and $\Theta = \mathbf{V}\Sigma(\mathbf{U}_{1:K,:})^\top$ gives

$$\begin{aligned} \begin{bmatrix} \mathbf{I}_r \\ \mathbf{W}_2^\top \end{bmatrix} \mathbf{C} &= \mathbf{V}\Sigma(\mathbf{U}_{1:r,:})^\top, \\ \begin{bmatrix} \mathbf{I}_r \\ \mathbf{W}_2^\top \end{bmatrix} \mathbf{C}\mathbf{W}_1 &= \mathbf{V}\Sigma(\mathbf{U}_{(r+1):N,:})^\top. \end{aligned} \tag{23}$$

Since $\mathbf{C}$ is full rank, $\mathbf{U}_{1:r,:}$ has to also be full rank and (23) can be translated into

$$\mathbf{U}_{(r+1):N,:} = \mathbf{W}_1^\top \mathbf{U}_{1:r,:}.$$

Therefore,

$$\mathbf{U}_{1:K,:} = \begin{bmatrix} \mathbf{I}_r \\ \mathbf{W}_1^\top \end{bmatrix} \mathbf{U}_{1:r,:}. \tag{24}$$

By plugging the (24) into (21) and again using the fact that $\mathbf{U}_{1:r,:}$ is full rank, we have

$$(\mathbf{\Pi} - \tilde{\mathbf{\Pi}}\mathbf{P}) \begin{bmatrix} \mathbf{I}_r \\ \mathbf{W}_1^\top \end{bmatrix} = \mathbf{0}. \tag{25}$$

Proof of part (b). Denote $\mathbf{A} := \mathbf{\Pi} - \tilde{\mathbf{\Pi}}\mathbf{P}$. If $r = K - 1$, $\mathbf{W}_1 = (W_{1,1}, \dots, W_{1,K-1})$ is a $(K - 1)$ -dimensional vector and (25) gives us

$$\mathbf{A}_{:,j} + W_{1,j}\mathbf{A}_{:,K} = 0, \quad \forall j = 1, \dots, K - 1. \tag{26}$$

Denote an r -dimensional column vector with all entries equal to one by $\mathbf{1}_r$. Right multiplying $\mathbf{1}_r$ to both sides of (25) yields

$$\mathbf{A} \begin{bmatrix} \mathbf{1}_r \\ \mathbf{W}_1^\top \mathbf{1}_r \end{bmatrix} = \mathbf{0}.$$

Also, note that both $\mathbf{\Pi}$ and $\tilde{\mathbf{\Pi}}\mathbf{P}$ have row sums of 1. Hence,

$$\begin{aligned} \sum_{j=1}^K \mathbf{A}_{:,j} &= \mathbf{0}_N, \\ \sum_{j=1}^{K-1} \mathbf{A}_{:,j} + \mathbf{W}_1^\top \mathbf{1}_r \mathbf{A}_{:,K} &= \mathbf{0}_N. \end{aligned}$$

Taking the difference of the two equations above gives $(1 - \mathbf{W}_1^\top \mathbf{1}_r) \mathbf{A}_{:,K} = \mathbf{0}_N$. If $\mathbf{W}_1^\top \mathbf{1}_r \neq 1$, then $\mathbf{A}_{:,K}$ has to be $\mathbf{0}_N$, which implies $\mathbf{A}_{:,j} = \mathbf{0}_N$ for all $j = 1 \dots, K-1$ according to (26). Therefore, $\mathbf{A} = \mathbf{\Pi} - \tilde{\mathbf{\Pi}}\mathbf{P} = \mathbf{0}_N$, which leads to $\tilde{\mathbf{\Theta}} = \mathbf{\Theta}\mathbf{P}^\top$.

Note that using (24) leads to

$$\mathbf{\Theta}^\top = \mathbf{U}_{1:K,:} \mathbf{\Sigma} \mathbf{V}^\top = \begin{bmatrix} \mathbf{I}_{K-1} \\ \mathbf{W}_1^\top \end{bmatrix} \mathbf{U}_{1:(K-1),:} \mathbf{\Sigma} \mathbf{V}^\top = \begin{bmatrix} \mathbf{I}_{K-1} \\ \mathbf{W}_1^\top \end{bmatrix} (\mathbf{\Theta}_{:,1:(K-1)})^\top.$$

Hence $\mathbf{\Theta}_{:,K} = \mathbf{\Theta}_{:,1:(K-1)} \mathbf{W}_1$. Therefore, the condition $\mathbf{W}_1^\top \mathbf{1}_r \neq 1$ is equivalent to the K -th column of $\mathbf{\Theta}$ not being an affine combination of the other columns.

Proof of part (c). Now consider the case of *either* $r = K-1$ with $\mathbf{W}_1^\top \mathbf{1}_r = 1$, *or* $r < K-1$. Assume subject m is completely mixed so that $\pi_{m,k} > 0, \forall k = 1, \dots, K$. Define

$$\tilde{\boldsymbol{\pi}}_i^\top = \begin{cases} \boldsymbol{\pi}_i^\top & \text{if } i \neq m \\ \boldsymbol{\pi}_m^\top + \epsilon \boldsymbol{\beta}^\top [-\mathbf{W}_1^\top, \mathbf{I}_{K-r}] & \text{if } i = m \end{cases},$$

where $\epsilon > 0$ is small enough so that $\tilde{\pi}_i \in (0, 1)$, and $\beta \in \mathbb{R}^{K-r}$ is such that $\beta^\top(\mathbf{1}_{K-r} - \mathbf{W}_1^\top \mathbf{1}_r) = 0$. Note that such $\beta \neq \mathbf{0}$ always exists under the assumption in part (c), because if $r = K - 1$ with $\mathbf{W}_1^\top \mathbf{1}_r = 1$, then $\beta^\top(\mathbf{1}_{K-r} - \mathbf{W}_1^\top \mathbf{1}_r) = \beta(1 - 1) = 0$ holds for any $\beta \in \mathbb{R}$; if $r < K - 1$, then $K - r \geq 2$ and β has dimension at least two, so the inner product equation $\beta^\top(\mathbf{1}_{K-r} - \mathbf{W}_1^\top \mathbf{1}_r) = 0$ must have a nonzero solution β . The constructed $\tilde{\Pi}$ have row sums of 1 by the construction of β . Furthermore, $\tilde{\Pi}\mathbf{U}_{1:K,:}$ and $\Pi\mathbf{U}_{1:K,:}$ can only be different on the m -th row, and

$$\tilde{\pi}_m^\top \mathbf{U}_{1:K,:} = \pi_m^\top \mathbf{U}_{1:K,:} + \epsilon \beta^\top [-\mathbf{W}_1^\top, \mathbf{I}_{K-r}] \begin{bmatrix} \mathbf{I}_r \\ \mathbf{W}_1^\top \end{bmatrix} \mathbf{U}_{1:r,:} = \pi_m^\top \mathbf{U}_{1:K,:}.$$

Hence $\tilde{\Pi}\mathbf{U}_{1:K,:} = \Pi\mathbf{U}_{1:K,:}$. This gives us

$$\Pi\Theta^\top = \Pi\mathbf{U}_{1:K,:}\Sigma\mathbf{V}^\top = \tilde{\Pi}\mathbf{U}_{1:K,:}\Sigma\mathbf{V}^\top = \tilde{\Pi}\Theta^\top.$$

We can see that (Π, Θ) and $(\tilde{\Pi}, \Theta)$ yield the same model but $\Pi \neq \tilde{\Pi}$. This completes the proof for part (c). $\square$

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